MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RADIATION COUNTER

TYPE 7616

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Low sensitivity, self-quenching for detection of beta and gamma radiation

See figure 1

Mounting position: Any

Weight: 0.4 ounce (11.34 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Rp	TA	Barometric pressure, reduced
Unit:	Meg	°C	mmHg
Maximum:	---	+55	87
Minimum:	---	-20	---
TEST CONDITIONS:	1.0	---	---

GENERAL:

Qualification - Required
Holding period - See note 3

Ⓕ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
6201	Background, contamination, and photosensitivity	1	t = 2 minutes (min)	0.65	II	N/t	---	10	Npm
6211	Starting voltage	2	Nps = 20 nominal; R1 = C1 = C2 = 0; R2 = 1 Meg; C3 = 0.01 µF	0.65	II	Es	710	---	Vdc
6221	Response count rate and current	4	50 r/hr	0.65	II	N/t	190	230	Nps
(F)6221	Response count rate (beta)	5,6		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
6226	Pulse amplitude	8	Nps = 20 nominal; R1 = 0.1 Meg; C1 = 0.01 µF	---	---	Ebb	---	760	Vdc
	Quality conformance inspection, part 3								
---	Life test	9	Group C; counting rate = 1,000 ±50 Nps; Ebb = 680 Vdc; t = 250 hours	---	---	---	---	---	---
---	Life-test end point	10		---	---	---	---	---	---
1031	Variable-frequency vibration	7,11, 13		---	---	---	---	---	---
6205	Leakage current	13	Ebb = 500 Vdc; relative humidity 90 ±5%	---	---	LIb	---	0.5	µAdc
---	Temperature	12,13	T = +50°C, -20°C	---	---	---	---	---	---
(F) 1041	Shock	7, 13	Hammer angle = 40°	---	---	---	---	---	---

NOTES:

1. The tube response (count rate) due to the combined effects of background, contamination, and photosensitivity determined as specified herein, shall be not more than the limit specified herein. The tube may be shielded from extraneous radiation fields by lead and aluminum interposed between it and the tube. The lead shall have a maximum thickness of 2 inches (50.8 mm), the aluminum shall have a maximum thickness of 1/8 inch (3.18 mm). In determining tube response (count rate), Ebb = 750 volts and Rp = 1 Meg. The count rates shall be determined using a scaler having a resolving time of 5 microseconds and a discrimination level of 0.25 volt.

2. Apply approximately 800 volts anode voltage. Decrease this voltage until the largest visible pulses are 2 volts ±20 percent. This voltage shall be considered the starting voltage.

3. All tubes shall pass this test. Testing shall be performed 30 or more days after previous production tests on the same tubes. A tube that fails this test may be retested after an additional 30 or more days with the provision that the value of Ebb for anode voltage for pulse amplitude test does not change by more than ±5 volts from the value at the beginning of the last 30-day period, and that the tubes conform to the initial limits for starting voltage and background, contamination, and photosensitivity.

4. Test shall be made using Government approved standard test unit. The source of radiation shall be cesium 137. (Preliminary calibration of unit shall be as directed on operating instructions.) The external connections shall be as follows:

   (a) Counting circuit shall have a discrimination of +3 volts.
   (b) DC high voltage input shall be 680 ±0.5 volts.
   (c) Trigger input shall be 1,000 ±10 prr.

5. Test shall be made with calibrated beta source. Ebb shall be anode voltage plus or minus 5 volts for 50-volt pulse as determined in the pulse amplitude test. The count rate shall be determined by using a scaler having a resolving time of 5 microseconds, and a discrimination level of 0.25 volt.

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NOTES: -Continued

6. The counting rate shall not vary by more than plus or minus 0.3 N/t from the value of N/t of the calibrated beta source furnished.

7. Criterion for passing this test shall be compliance after test of at least 80 percent of the tubes with the requirements for the following:

   (a) Starting voltage.
   (b) Pulse amplitude.
   (c) Response count rate (beta).

8. Test shall be conducted in the circuit as shown on figure 2 (in lieu of figure 6211-1. The procedure shall be as follows:

   Apply 950 volts anode voltage. Observe the pulse pattern to determine if the tube goes into glow discharge. If the tube goes into a glow discharge, it shall be considered as a defective tube. If no glow discharge occurs, decrease this voltage until the largest visible pulses are 50 ±2 volts. This voltage shall be considered anode voltage for a 50 volt pulse.

9. The tube shall be placed in a radiation field that will produce a counting rate of 1,000 ±50 Nps when it is activated with a voltage pulse that has a maximum width of 50 µs, a nominal height of 150 volts and a prr of 1,400 ±200.

10. The limits for acceptability shall be as follows:

    (a) Background, contamination, and photosensitivity: Initial limits specified herein.
    (b) Starting voltage: 695 volts minimum.
    (c) Anode voltage for 50 volt pulse: Initial value -15, +5 volts.

11. The tube shall be mounted with its longitudinal axis horizontal and perpendicular to the direction of motion by means of a rigid clamp at the center of the cathode. The time for gradually covering the frequency range shall be 15 minutes.

12. The functional requirements of the tube over the stipulated range of temperature conditions shall be as follows:

    (a) Background, contamination, and photosensitivity: Initial limits specified herein.
    (b) Starting voltage: Initial limits specified herein.
    (c) Pulse amplitude: Initial value ±6 volts at +50°C, ±12 volts at -20°C.

(F)13. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

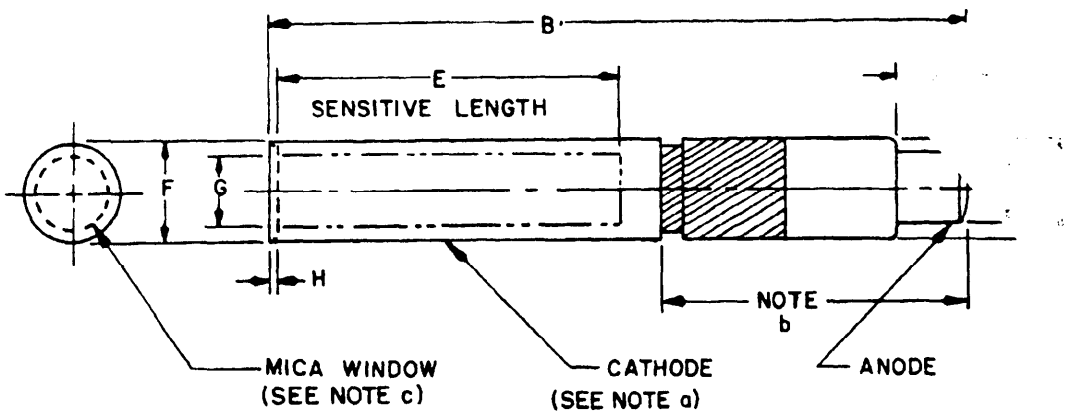
User activities:
Army - ME, WC, MU
Navy - AS, OS, MC

Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	.328 (8.33)	.359 (9.12)
B	2.469 (62.71)	2.531 (64.29)
C	.245 (6.22)	.255 (6.48)
E		1.250 (31.75)
F	.339 (8.61)	.349 (8.86)
G	.248 (6.30)	.252 (6.40)
H		.020 (.51)
Reference dimension		
D	.250 (6.35)	

NOTES:
a. Cathode material - Allegheny Ludium Sealmet 1 steel for glass to metal seals, or equal.
b. Mechanical configuration and materials used may vary, provided all other specified dimensions are met.
c. Mica window: 3 to 4 mg/cm^2.

FIGURE 1. Outline drawing of electron tube type 7616.

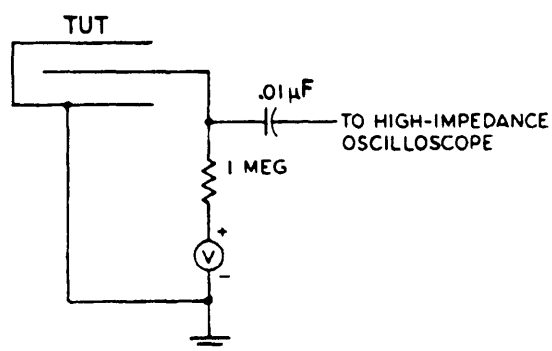


FIGURE 2. Circuit for measuring starting voltage and 50 volt pulse height.

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